

Abstract: This study investigates the role of information theory in designing cryptographic protocols that establish trust in adversarial settings. By leveraging secure multi-party computations, differential privacy, and zero-knowledge proofs, we aim to develop low-level concurrent, open-source secure computer code—Adversarial Trust Framework (ATF)—demonstrating these results.

Introduction: Adversarial cooperation seeks to mitigate unintended effects of conflict by structuring secure mechanisms for interaction.

This research, in collaboration with the Chinese Academy of Computer Sciences, aims to develop cryptographic frameworks designed to alleviate conflict through information-secure, verifiable encryption protocols.

Research Questions:

What does it mean to apply cryptography to cooperation?

How can the proposed cryptographic protocols be improved through interdisciplinary collaboration?

Research Objectives:

Demonstrate an Adversarial Trust Framework for classical game-theoretic problems.

Open-source a software library based on this research.

Literature Review:

Cryptography extends far beyond encryption serves as the foundation of digital trust. Zero-knowledge proofs, homomorphic encryption, and multi-party computation enable collaborative problem-solving while maintaining individual privacy.

Methodology: The study will focus on the following structured games:

Playing card games without revealing one's cards: Demonstrate how to claim victory in a card game without exposing the cards at hand.

Playing tic-tac-toe without revealing one's moves: Demonstrate how a player can prove they have a winning strategy in tic-tac-toe without disclosing the actual moves.

Classical adversarial stag-hunt: Implement an Adversarial Trust Framework for this fundamental cooperation game.

Expected Outcomes:

Development of Adversarial Trust Framework (ATF), an open-source cryptographic protocol suite enabling cooperation in adversarial settings.

Demonstration of privacy-preserving game-theoretic solutions.

Improved understanding of how cryptography can facilitate trust in conflict.

Research Timeline:

Month 1-8: Literature review and foundational learning.

Month 9-15: Protocol design, coding, and experimentation.

Month 15-24: Analysis, validation, and final reporting.

Conclusion: This research seeks to advance cryptographic techniques that promote cooperation among adversaries while ensuring strategic privacy. By applying secure computation and cryptographic proofs, we aim to develop practical solutions that contribute to peace-building and trust establishment in conflict-prone environments.

References: (To be completed with relevant academic sources on cryptographic protocols, adversarial game theory, and secure computation.)